assignment_02_key

Bill Perry

Load Libraries

```
# load libraries
library(skimr)
library(janitor)
```

```


Attaching package: 'janitor'
```

```
The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```
library(knitr)
library(patchwork)
library(corrplot)     # For correlation plots
```

```
corrplot 0.95 loaded
```

```
library(GGally)       # For pairs plots
```

```
Loading required package: ggplot2
```

```
library(broom)        # For tidy model outputs
library(performance)  # For model performance metrics
library(see)          # For better diagnostic plots
library(car)
```

```
Loading required package: carData
```

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.1
✔ lubridate 1.9.4     ✔ tibble    3.3.0
✔ purrr     1.1.0     ✔ tidyr     1.3.1
```

```
── Conflicts ──────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
✖ dplyr::recode() masks car::recode()
✖ purrr::some()   masks car::some()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Load dataframe

```
lake_df <- read_csv("data/cheruvelil_lake_data.csv") %>% clean_names()
```

```
Rows: 593 Columns: 16
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (7): lake, state, source, state_lake_id, epa_lake_id, date, AgUrb.prop
dbl (9): longitude_utm, latitude_utm, year, CHL_ugL, TN_ugL, TP_ugL, Area_km...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Fix the watershed proportion

```
lake_df <- lake_df %>%
  mutate(watershed_proportion = as.numeric(if_else(ag_urb_prop == "low", 0,1)))

lake_df <- lake_df %>% select(-ag_urb_prop)
```

Make variable transformation

```
lake_df <- lake_df %>%
  mutate(
    # Log-transform the response variable and key predictors
    log_chl_ug_l      = log10(chl_ug_l),          # Log chlorophyll (response)
    log_tn_ug_l       = log10(tn_ug_l),           # Log total nitrogen
    log_tp_ug_l       = log10(tp_ug_l),           # Log total phosphorus
    log_area_km2      = log10(area_km2),          # Log lake area
    log_z_mean_m      = log10(z_mean_m),          # Log mean depth
    log_watershed_km2 = log10(watershed_km2))
```

Make long format dataframe

```
# lets make a long formate dataframe for easy graphing
lake_long_df <- lake_df %>%
  pivot_longer(cols = -c(lake, longitude_utm, latitude_utm, state, source,
                         state_lake_id, epa_lake_id, date, year),
               names_to = "variable",
               values_to = "value")
```

Look at data

```
head(lake_df)
```

```
# A tibble: 6 × 22
  lake   longitude_utm latitude_utm state source state_lake_id epa_lake_id date
  <chr>          <dbl>        <dbl> <chr> <chr>  <chr>         <chr>       <chr>
1 Lacey…       337515.     1973360. IA    IADNR… 50            IA050       8/4/…
2 Lake …       385681.     1987587. IA    IADNR… 55            IA055       8/3/…
3 Alcon…       962961.     2459749. MI    MI_ST… 1_4           MI1_4       8/25…
4 Lake …       801078.     2457101. MI    MI_DN… 10_111        MI10_111    <NA>
5 Lower…       774320.     2437134. MI    MI_DN… 10_24         MI10_24     <NA>
6 Platt…       782376.     2451889. MI    MI_ST… 10_39         MI10_39     8/12…
# ℹ 14 more variables: year <dbl>, chl_ug_l <dbl>, tn_ug_l <dbl>,
#   tp_ug_l <dbl>, area_km2 <dbl>, z_mean_m <dbl>, watershed_km2 <dbl>,
#   watershed_proportion <dbl>, log_chl_ug_l <dbl>, log_tn_ug_l <dbl>,
#   log_tp_ug_l <dbl>, log_area_km2 <dbl>, log_z_mean_m <dbl>,
#   log_watershed_km2 <dbl>
```

```r
lake_df %>% skim()
```

Name	Piped data
Number of rows	593
Number of columns	22
_______________________	
Column type frequency:	
character	6
numeric	16
________________________	
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
lake	0	1.00	7	36	0	510	0
state	0	1.00	2	2	0	3	0
source	0	1.00	9	13	0	5	0
state_lake_id	0	1.00	1	8	0	593	0
epa_lake_id	0	1.00	5	10	0	593	0
date	78	0.87	6	7	0	168	0

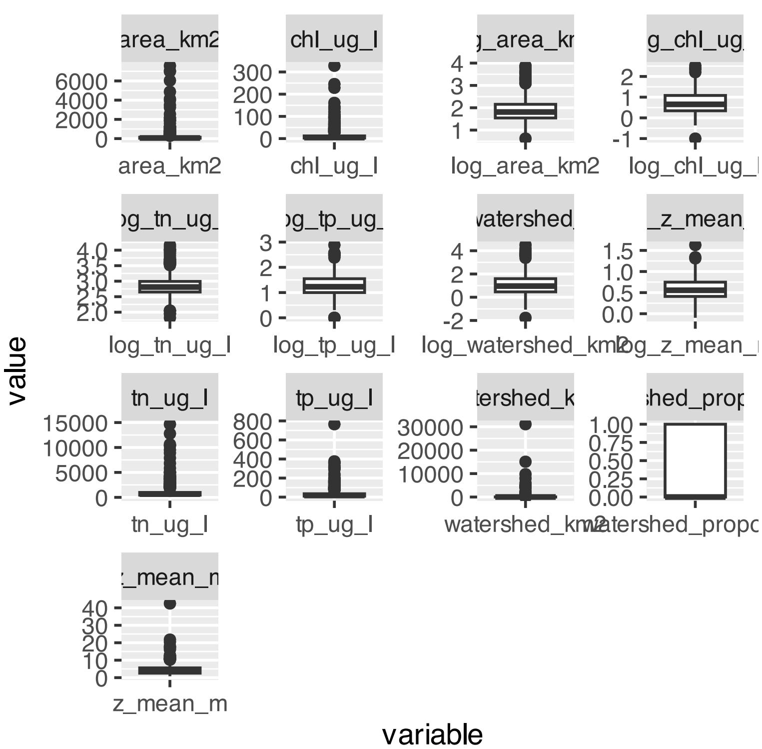
Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
longitude_utm	0	1	643989.35	282496.03	−37508.79	436118.90	772612.27	862228.33	1048083.47	
latitude_utm	0	1	2349913.86	198036.63	1956356.03	2182084.72	2345964.32	2517187.79	2745426.16	
year	0	1	1990.45	12.13	1975.00	1980.00	1981.00	2004.00	2004.00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chl_ug_l	0	1	14.89	29.35	0.10	2.20	4.50	12.10	327.84	
tn_ug_l	0	1	1052.63	1441.12	66.00	447.00	648.00	984.00	14661.00	
tp_ug_l	0	1	38.18	62.71	1.00	10.00	17.00	35.32	765.00	
area_km2	0	1	226.92	710.38	4.18	34.91	64.56	142.79	7575.07	
z_mean_m	0	1	4.46	3.16	0.80	2.57	3.60	5.60	42.50	
watershed_km2	0	1	244.74	1614.71	0.02	2.89	9.12	39.38	31079.86	
watershed_proportion	0	1	0.34	0.47	0.00	0.00	0.00	1.00	1.00	
log_chl_ug_l	0	1	0.75	0.57	−1.00	0.34	0.65	1.08	2.52	
log_tn_ug_l	0	1	2.86	0.33	1.82	2.65	2.81	2.99	4.17	
log_tp_ug_l	0	1	1.30	0.45	0.00	1.00	1.23	1.55	2.88	
log_area_km2	0	1	1.88	0.54	0.62	1.54	1.81	2.15	3.88	
log_z_mean_m	0	1	0.57	0.25	−0.10	0.41	0.56	0.75	1.63	
log_watershed_km2	0	1	1.10	0.87	−1.77	0.46	0.96	1.60	4.49	

Lets do boxplots of all variables

```
lake_long_df %>%
  ggplot(aes(variable, value)) + geom_boxplot() + facet_wrap(~variable, scales = "free")
```



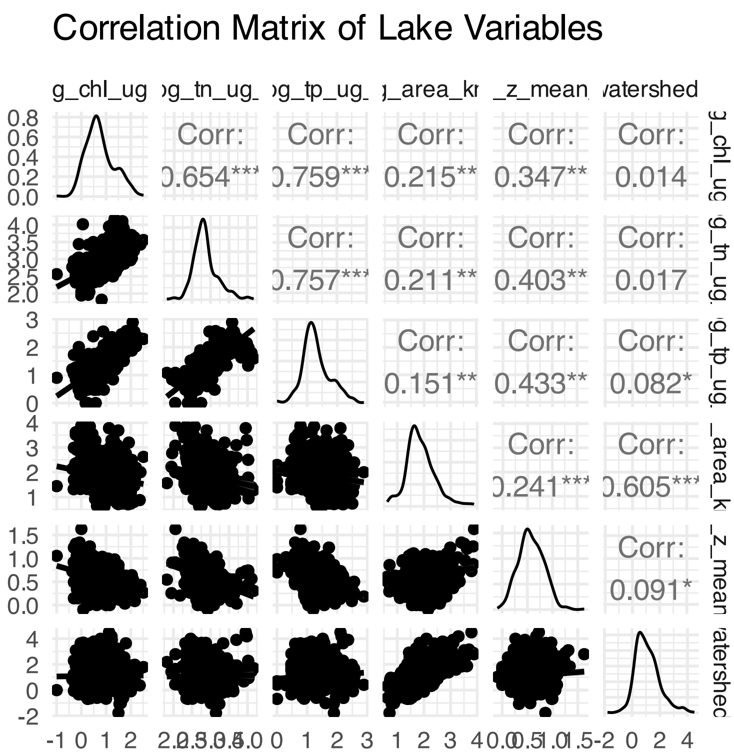
Look at correlation matrix plot of all data

```r
# Create correlation matrix for numeric variables only
numeric_vars <- lake_df %>%
  select(
  log_chl_ug_l       ,
  log_tn_ug_l        ,
  log_tp_ug_l        ,
  log_area_km2       ,
  log_z_mean_m       ,
  log_watershed_km2
  )

# Calculate correlation matrix
cor_matrix <- cor(numeric_vars, use = "complete.obs")

# Display correlation matrix
round(cor_matrix, 3)
```

```
                  log_chl_ug_l log_tn_ug_l log_tp_ug_l log_area_km2
log_chl_ug_l             1.000       0.654       0.759       -0.215
log_tn_ug_l              0.654       1.000       0.757       -0.211
log_tp_ug_l              0.759       0.757       1.000       -0.151
log_area_km2            -0.215      -0.211      -0.151        1.000
log_z_mean_m            -0.347      -0.403      -0.433        0.241
log_watershed_km2        0.014       0.017       0.082        0.605
                  log_z_mean_m log_watershed_km2
log_chl_ug_l            -0.347             0.014
log_tn_ug_l             -0.403             0.017
log_tp_ug_l             -0.433             0.082
log_area_km2             0.241             0.605
log_z_mean_m             1.000             0.091
log_watershed_km2        0.091             1.000
```

Look at the correlations another way

```r
# Create ggpairs plot
ggpairs(numeric_vars,
        lower = list(continuous = "smooth"),    # Scatterplots with smooth lines
        upper = list(continuous = "cor"),       # Correlation coefficients
        diag = list(continuous = "density"),    # Density plots on diagonal
        title = "Correlation Matrix of Lake Variables") +
  theme_minimal()
```



Start the multiple regression

Intial Multiple Regression# Start the multiple regression

```r
# Fit initial model with all predictors (Model 1)
model_initial <- lm(chl_ug_l ~ tn_ug_l + tp_ug_l + area_km2 +
                z_mean_m + watershed_km2 + watershed_proportion , data = lake_df)

# Get model summary
summary(model_initial)
```

```

Call:
lm(formula = chl_ug_l ~ tn_ug_l + tp_ug_l + area_km2 + z_mean_m +
    watershed_km2 + watershed_proportion, data = lake_df)

Residuals:
     Min       1Q   Median       3Q      Max
-108.726   -5.285   -1.575    2.336  257.146

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)           2.443838   1.935939   1.262 0.207324
tn_ug_l               0.002850   0.000708   4.026 6.42e-05 ***
tp_ug_l               0.272648   0.016078  16.958  < 2e-16 ***
area_km2              0.002132   0.001555   1.371 0.170817
z_mean_m             -0.614960   0.334853  -1.837 0.066789 .
watershed_km2        -0.002086   0.000612  -3.409 0.000697 ***
watershed_proportion  5.260065   2.010798   2.616 0.009128 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
Residual standard error: 20.95 on 586 degrees of freedom
Multiple R-squared:  0.4958,    Adjusted R-squared:  0.4906
F-statistic: 96.03 on 6 and 586 DF,  p-value: < 2.2e-16
```

```r
# Calculate Variance Inflation Factors (VIF)
vif_initial_values <- vif(model_initial)
print("Variance Inflation Factors:")
```

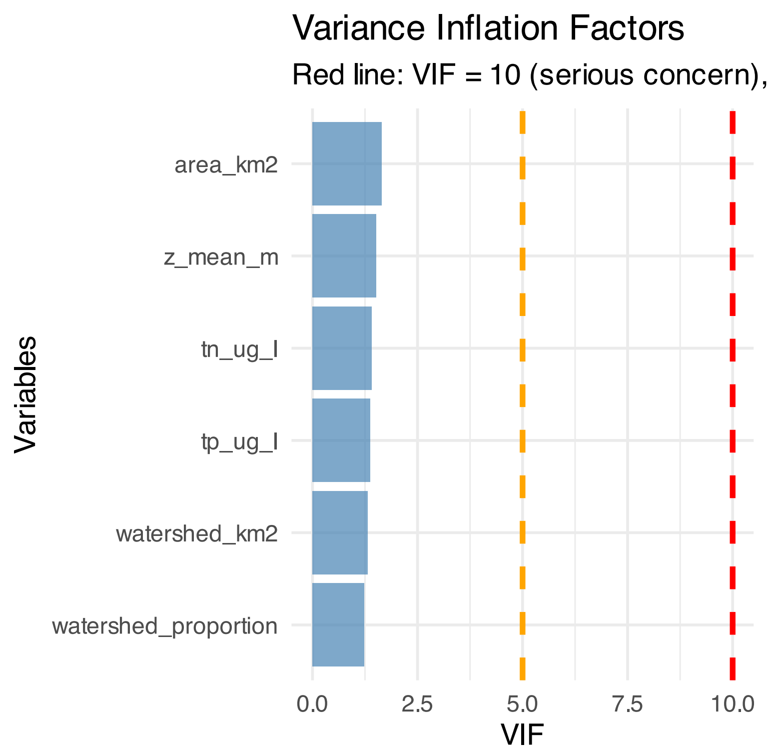
```
[1] "Variance Inflation Factors:"
```

```r
print(vif_initial_values)
```

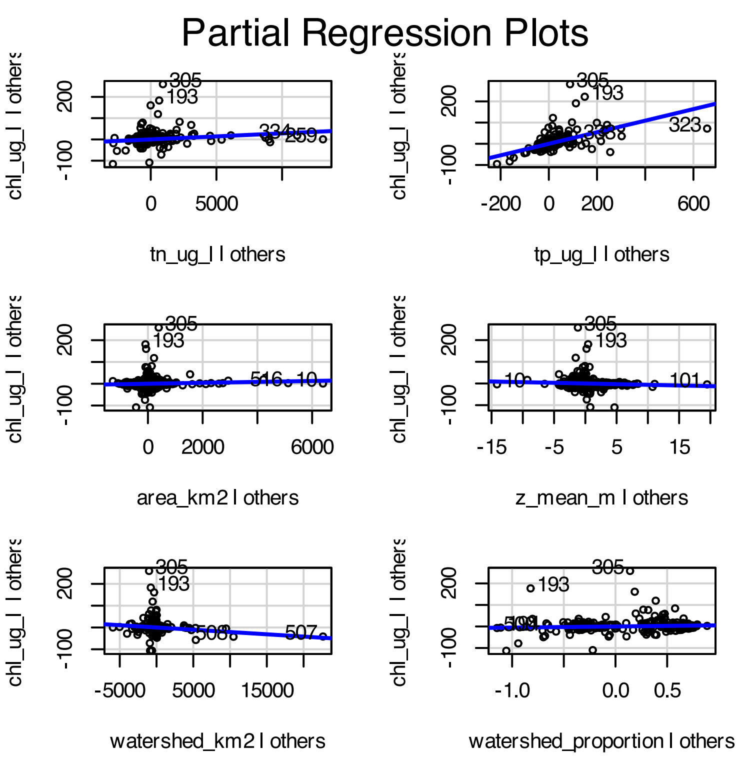
```
             tn_ug_l              tp_ug_l            area_km2
            1.404011             1.370824            1.645615
            z_mean_m        watershed_km2 watershed_proportion
            1.510612             1.316867            1.229766
```

VIF Visual cmparison

```r
# Create a data frame for better visualization
vif_initial_df <- data.frame(
  Variable = names(vif_initial_values),
  VIF = as.numeric(vif_initial_values)
) %>%
  arrange(desc(VIF))

# Visualize VIF values
ggplot(vif_initial_df, aes(x = reorder(Variable, VIF), y = VIF)) +
  geom_col(fill = "steelblue", alpha = 0.7) +
  geom_hline(yintercept = 10, color = "red", linetype = "dashed",
             linewidth = 1) +
  geom_hline(yintercept = 5, color = "orange", linetype = "dashed",
             linewidth = 1) +
  coord_flip() +
  labs(
    title = "Variance Inflation Factors",
    subtitle = "Red line: VIF = 10 (serious concern), Orange line: VIF = 5 (moderate concern)",
    x = "Variables",
    y = "VIF"
  ) +
  theme_minimal()
```

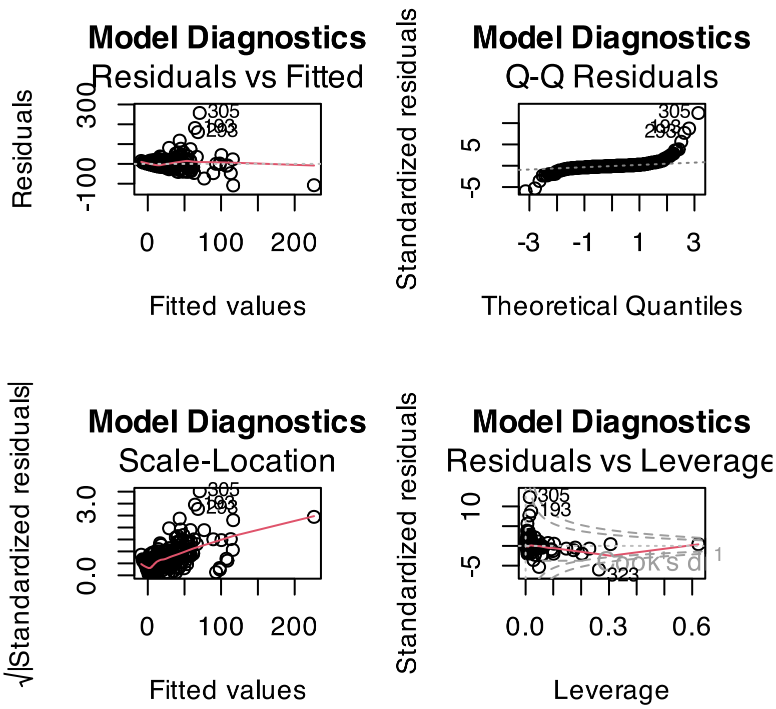



```r
# Create partial regression plots (Added Variable Plots)
# This helps us see the relationship between each predictor and response
# after accounting for other variables

par(mfrow = c(2, 3))
avPlots(model_initial, main = "Partial Regression Plots")
```



```r
par(mfrow = c(1, 1))
```

```r
# Create diagnostic plots
par(mfrow = c(2, 2))
plot(model_initial, main = "Model Diagnostics")
```




```r
par(mfrow = c(1, 1))
```

Multiple Regression wiht logged response variable

```r
# Fit initial model with all predictors (Model 1)
model_log_response <- lm(log_chl_ug_l ~ tn_ug_l + tp_ug_l + area_km2 +
                z_mean_m + watershed_km2 + watershed_proportion , data = lake_df)

# Get model summary
summary(model_log_response)
```

```

Call:
lm(formula = log_chl_ug_l ~ tn_ug_l + tp_ug_l + area_km2 + z_mean_m +
    watershed_km2 + watershed_proportion, data = lake_df)

Residuals:
     Min       1Q   Median       3Q      Max
-1.94887 -0.26553  0.01545  0.29255  1.00743

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)
(Intercept)           5.943e-01  3.754e-02  15.832  < 2e-16 ***
tn_ug_l               7.195e-05  1.373e-05   5.241 2.24e-07 ***
```

```
tp_ug_l               3.868e-03  3.118e-04  12.408  < 2e-16 ***
area_km2              2.498e-05  3.015e-05   0.829    0.408    
z_mean_m             -3.388e-02  6.493e-03  -5.218 2.51e-07 ***
watershed_km2        -1.901e-05  1.187e-05  -1.602    0.110    
watershed_proportion  2.453e-01  3.899e-02   6.291 6.18e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4063 on 586 degrees of freedom
Multiple R-squared:  0.4959,	Adjusted R-squared:  0.4907 
F-statistic: 96.07 on 6 and 586 DF,  p-value: < 2.2e-16
```

```
# Calculate Variance Inflation Factors (VIF)
vif_log_resp_values <- vif(model_log_response)
print("Variance Inflation Factors:")
```

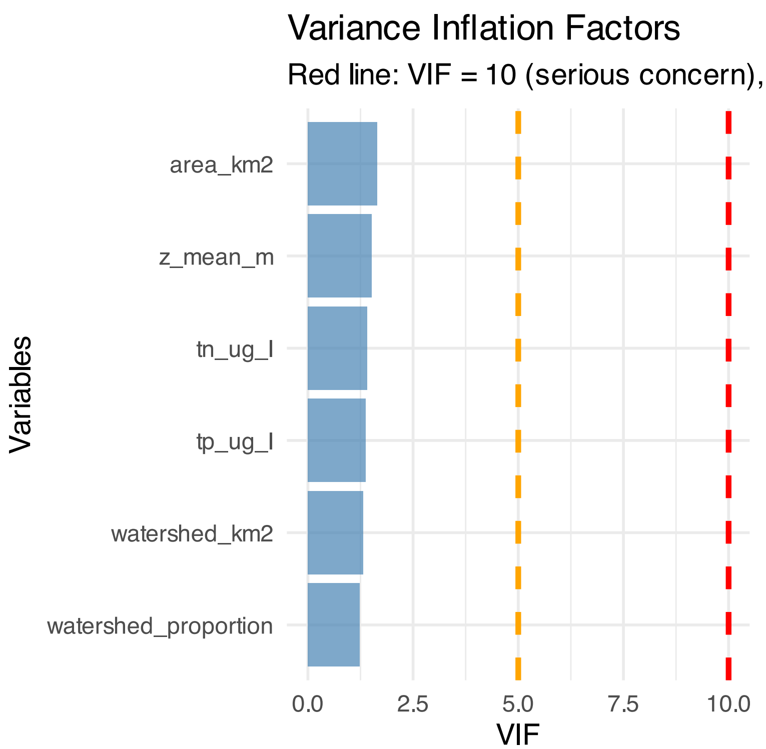
```
[1] "Variance Inflation Factors:"
```

```
print(vif_log_resp_values)
```

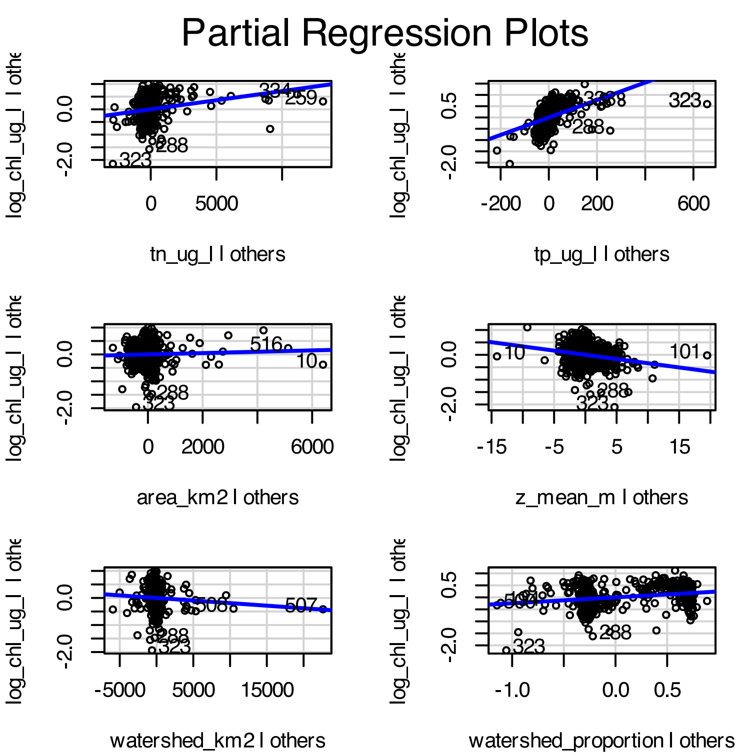
```
             tn_ug_l              tp_ug_l             area_km2 
            1.404011             1.370824             1.645615 
            z_mean_m        watershed_km2 watershed_proportion 
            1.510612             1.316867             1.229766 
```

VIF Visual cmparison

```
# Create a data frame for better visualization
vif_log_response_df <- data.frame(
  Variable = names(vif_log_resp_values),
  VIF = as.numeric(vif_log_resp_values)
) %>%
  arrange(desc(VIF))

# Visualize VIF values
ggplot(vif_log_response_df, aes(x = reorder(Variable, VIF), y = VIF)) +
  geom_col(fill = "steelblue", alpha = 0.7) +
  geom_hline(yintercept = 10, color = "red", linetype = "dashed",
             linewidth = 1) +
  geom_hline(yintercept = 5, color = "orange", linetype = "dashed",
             linewidth = 1) +
  coord_flip() +
  labs(
    title = "Variance Inflation Factors",
    subtitle = "Red line: VIF = 10 (serious concern), Orange line: VIF = 5 (moderate concern)",
    x = "Variables",
    y = "VIF"
  ) +
  theme_minimal()
```

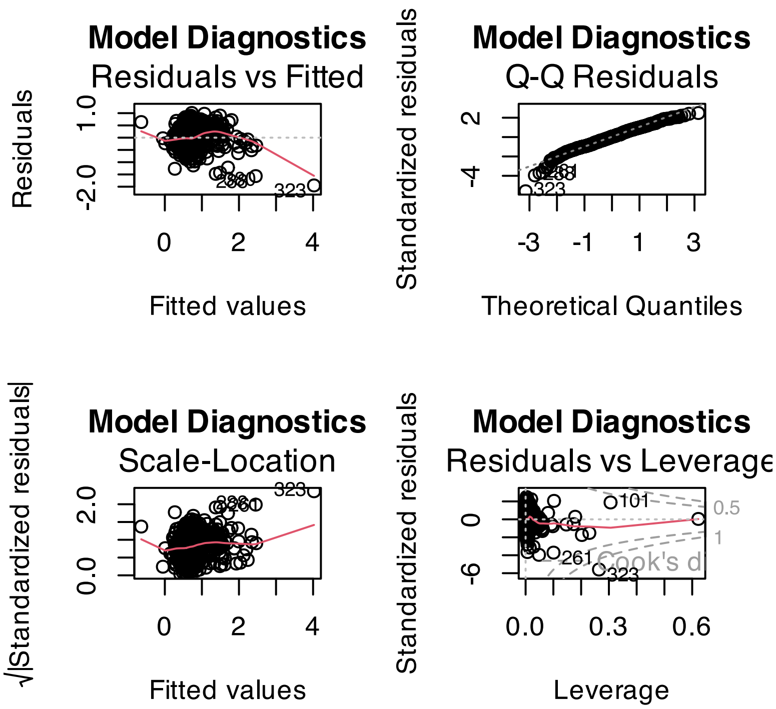



```r
# Create partial regression plots (Added Variable Plots)
# This helps us see the relationship between each predictor and response
# after accounting for other variables

par(mfrow = c(2, 3))
avPlots(model_log_response, main = "Partial Regression Plots")
```



```r
par(mfrow = c(1, 1))
```

```r
# Create diagnostic plots
par(mfrow = c(2, 2))
plot(model_log_response, main = "Model Diagnostics")
```




```r
par(mfrow = c(1, 1))
```

Note log response helps a LOT with normality and fan shaped residulas

Multiple Regression of ALL except watershed_proportion

```r
# Fit initial model with all predictors (Model 1)
model_log_all <- lm(log_chl_ug_l ~ log_tn_ug_l + log_tp_ug_l + log_area_km2 +
                log_z_mean_m + log_watershed_km2 + watershed_proportion , data = lake_df)

# Get model summary
summary(model_log_all)
```

```

Call:
lm(formula = log_chl_ug_l ~ log_tn_ug_l + log_tp_ug_l + log_area_km2 +
    log_z_mean_m + log_watershed_km2 + watershed_proportion,
    data = lake_df)

Residuals:
     Min       1Q   Median       3Q      Max
-1.60510 -0.19614  0.03072  0.23404  0.96053

```

```
Coefficients:
                      Estimate Std. Error t value Pr(>|t|)    
(Intercept)          -0.814032   0.193323  -4.211 2.95e-05 ***
log_tn_ug_l           0.257695   0.070913   3.634 0.000304 ***
log_tp_ug_l           0.749803   0.053583  13.993  < 2e-16 ***
log_area_km2         -0.104033   0.036300  -2.866 0.004307 ** 
log_z_mean_m          0.004493   0.068245   0.066 0.947529    
log_watershed_km2     0.016205   0.021943   0.738 0.460513    
watershed_proportion  0.077600   0.036939   2.101 0.036087 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3614 on 586 degrees of freedom
Multiple R-squared:  0.601, Adjusted R-squared:  0.5969 
F-statistic: 147.1 on 6 and 586 DF,  p-value: < 2.2e-16
```

```r
# Calculate Variance Inflation Factors (VIF)
vif_log_all_values <- vif(model_log_all)
print("Variance Inflation Factors:")
```

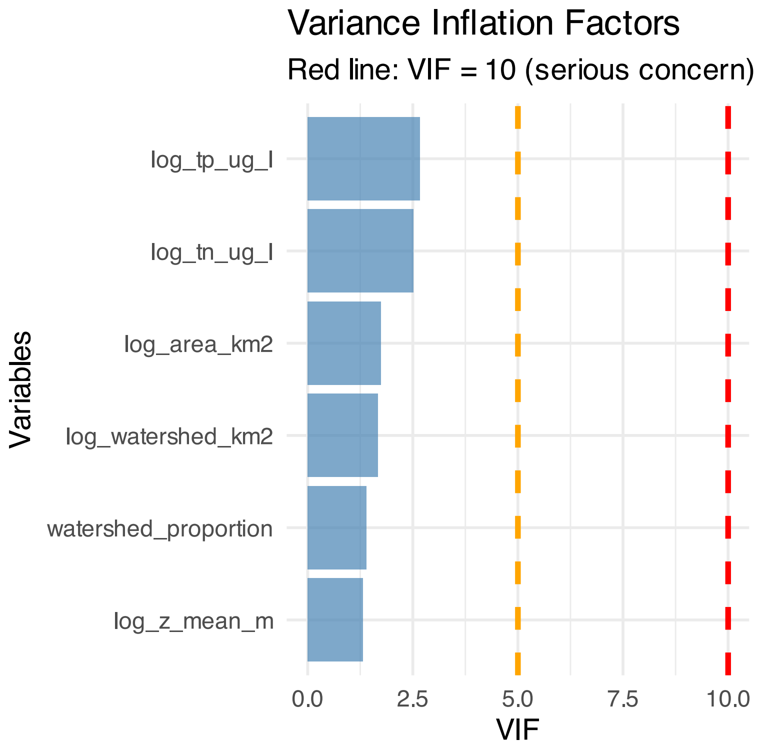
```
[1] "Variance Inflation Factors:"
```

```r
print(vif_log_all_values)
```

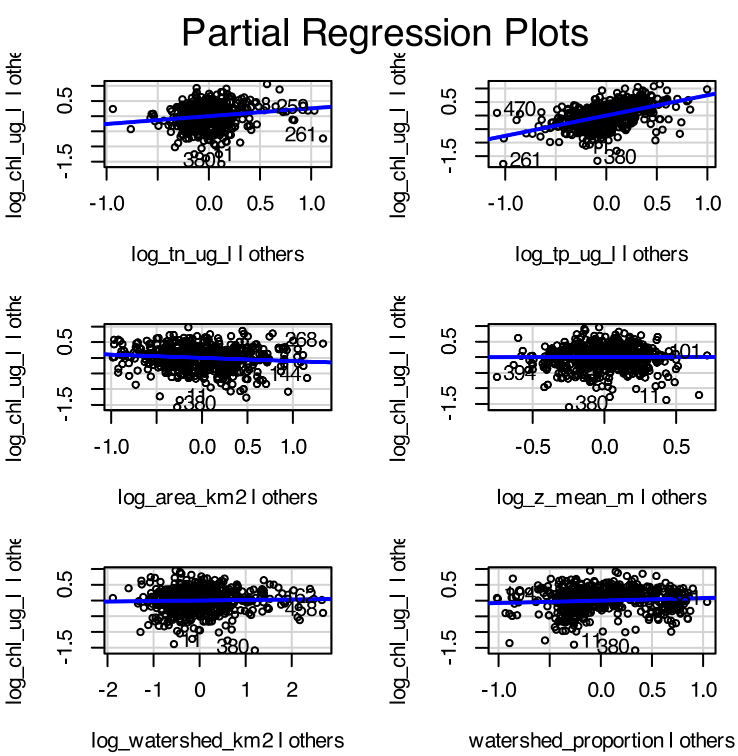
```
         log_tn_ug_l          log_tp_ug_l         log_area_km2 
            2.518023             2.674049             1.747897 
        log_z_mean_m    log_watershed_km2 watershed_proportion 
            1.319573             1.667116             1.394434 
```

VIF Visual cmparison

```r
# Create a data frame for better visualization
vif_log_all_df <- data.frame(
  Variable = names(vif_log_all_values),
  VIF = as.numeric(vif_log_all_values)
) %>%
  arrange(desc(VIF))

# Visualize VIF values
ggplot(vif_log_all_df, aes(x = reorder(Variable, VIF), y = VIF)) +
  geom_col(fill = "steelblue", alpha = 0.7) +
  geom_hline(yintercept = 10, color = "red", linetype = "dashed",
             linewidth = 1) +
  geom_hline(yintercept = 5, color = "orange", linetype = "dashed",
             linewidth = 1) +
  coord_flip() +
  labs(
    title = "Variance Inflation Factors",
    subtitle = "Red line: VIF = 10 (serious concern), Orange line: VIF = 5 (moderate concern)",
    x = "Variables",
    y = "VIF"
  ) +
  theme_minimal()
```

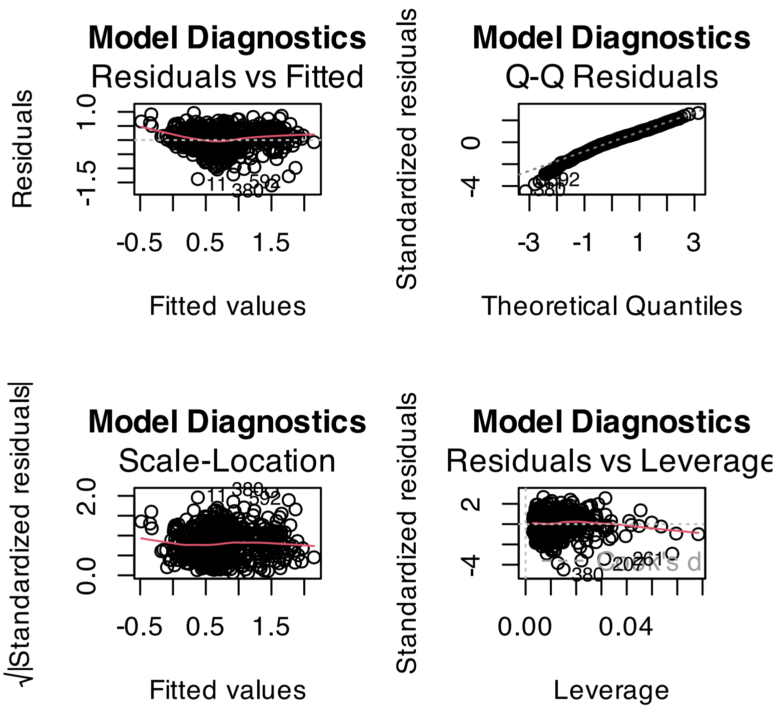



```r
# Create partial regression plots (Added Variable Plots)
# This helps us see the relationship between each predictor and response
# after accounting for other variables

par(mfrow = c(2, 3))
avPlots(model_log_all, main = "Partial Regression Plots")
```



```r
par(mfrow = c(1, 1))
```

```r
# Create diagnostic plots
par(mfrow = c(2, 2))
plot(model_log_all, main = "Model Diagnostics")
```




```r
par(mfrow = c(1, 1))
```

Note log of all is ok too but makes it hard to interpret the data…

compare the AIC of all the models

```r
# Quick diagnostic comparison
cat("Initial response AIC:", round(AIC(model_initial), 2), "\n")
```

```
Initial response AIC: 5299.87
```

```r
cat("Log response AIC:", round(AIC(model_log_response), 2), "\n")
```

```
Log response AIC: 623.49
```

```r
cat("Log all AIC:", round(AIC(model_log_all), 2), "\n")
```

```
Log all AIC: 484.87
```

AIC is way better for log respinse and a bit better for log all but still messy…

Model Simplification and Comparison

Remove Non-significant Variables

So this means removing area_km2 and watershed_km2 and lets see where we are

```
# # Fit initial model with all predictors (Model 1)
# model_log_response <- lm(log_chl_ug_l ~ tn_ug_l + tp_ug_l + area_km2 +
#                  z_mean_m + watershed_km2 + watershed_proportion , data = lake_df)

# Get model summary
summary(model_log_response)
```

```

Call:
lm(formula = log_chl_ug_l ~ tn_ug_l + tp_ug_l + area_km2 + z_mean_m +
    watershed_km2 + watershed_proportion, data = lake_df)

Residuals:
     Min       1Q   Median       3Q      Max
-1.94887 -0.26553  0.01545  0.29255  1.00743

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)
(Intercept)           5.943e-01  3.754e-02  15.832  < 2e-16 ***
tn_ug_l               7.195e-05  1.373e-05   5.241 2.24e-07 ***
tp_ug_l               3.868e-03  3.118e-04  12.408  < 2e-16 ***
area_km2              2.498e-05  3.015e-05   0.829    0.408
z_mean_m             -3.388e-02  6.493e-03  -5.218 2.51e-07 ***
watershed_km2        -1.901e-05  1.187e-05  -1.602    0.110
watershed_proportion  2.453e-01  3.899e-02   6.291 6.18e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4063 on 586 degrees of freedom
Multiple R-squared:  0.4959,	Adjusted R-squared:  0.4907
F-statistic: 96.07 on 6 and 586 DF,  p-value: < 2.2e-16
```

Simplified Model

```
model_log_response_reduced <- lm(log_chl_ug_l ~ tn_ug_l + tp_ug_l +
                z_mean_m + watershed_proportion , data = lake_df)

# Get model summary
summary(model_log_response_reduced)
```

```

Call:
lm(formula = log_chl_ug_l ~ tn_ug_l + tp_ug_l + z_mean_m + watershed_proportion,
    data = lake_df)

Residuals:
     Min       1Q   Median       3Q      Max
-1.94094 -0.26415  0.01817  0.29139  1.01224

```

```
Coefficients:
                      Estimate Std. Error t value Pr(>|t|)    
(Intercept)          5.876e-01  3.606e-02  16.293  < 2e-16 ***
tn_ug_l              6.661e-05  1.315e-05   5.065 5.46e-07 ***
tp_ug_l              3.883e-03  3.105e-04  12.506  < 2e-16 ***
z_mean_m            -3.156e-02  5.481e-03  -5.758 1.37e-08 ***
watershed_proportion  2.524e-01  3.845e-02   6.565 1.15e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4065 on 588 degrees of freedom
Multiple R-squared:  0.4936,	Adjusted R-squared:  0.4902 
F-statistic: 143.3 on 4 and 588 DF,  p-value: < 2.2e-16
```

Ok now we have cool results of how all of these affect the model and the model looks good…

Get a tidy model

```
final_summary <- tidy(model_log_response_reduced, conf.int = TRUE)
final_summary
```

```
# A tibble: 5 × 7
  term                 estimate std.error statistic  p.value conf.low conf.high
  <chr>                   <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept)           5.88e-1 0.0361        16.3  1.57e-49  5.17e-1   6.58e-1
2 tn_ug_l               6.66e-5 0.0000131      5.07 5.46e- 7  4.08e-5   9.24e-5
3 tp_ug_l               3.88e-3 0.000310      12.5  5.51e-32  3.27e-3   4.49e-3
4 z_mean_m             -3.16e-2 0.00548       -5.76 1.37e- 8 -4.23e-2  -2.08e-2
5 watershed_proportion  2.52e-1 0.0384         6.56 1.15e-10  1.77e-1   3.28e-1
```

Calculate partial R-squared for each variable

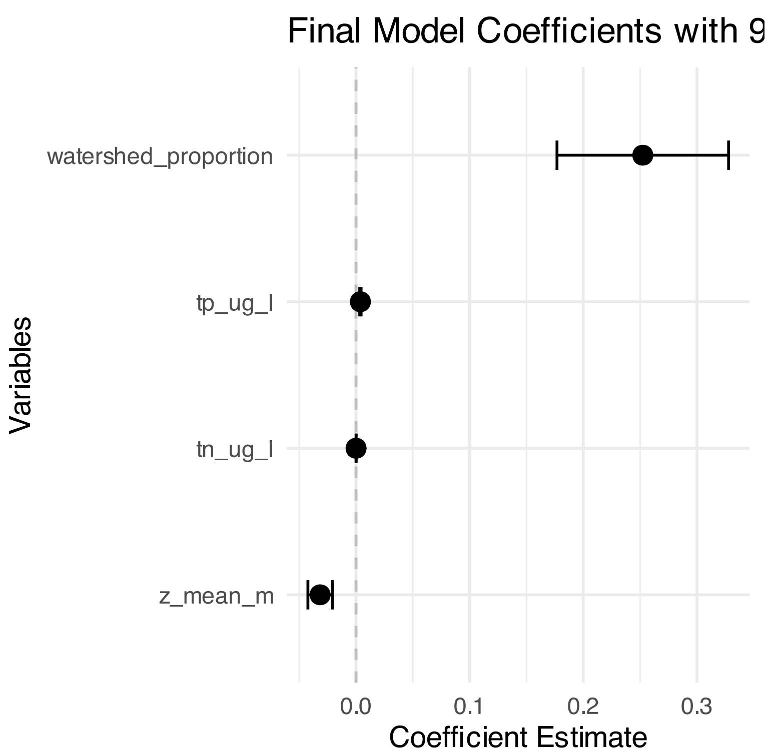
```
anova_table <- anova(model_log_response_reduced)
print("ANOVA Table:")
```

```
[1] "ANOVA Table:"
```

```
print(anova_table)
```

```
Analysis of Variance Table

Response: log_chl_ug_l
                      Df Sum Sq Mean Sq F value    Pr(>F)    
tn_ug_l                1 39.450  39.450 238.784 < 2.2e-16 ***
tp_ug_l                1 42.310  42.310 256.092 < 2.2e-16 ***
z_mean_m               1  5.818   5.818  35.218 5.039e-09 ***
watershed_proportion   1  7.120   7.120  43.094 1.147e-10 ***
Residuals            588 97.145   0.165                      
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Create coefficient plot
final_summary %>%
  filter(term != "(Intercept)") %>%
  mutate(term = fct_reorder(term, estimate)) %>%
  ggplot(aes(x = estimate, y = term)) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray") +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2) +
  labs(
    title = "Final Model Coefficients with 95% Confidence Intervals",
    x = "Coefficient Estimate",
    y = "Variables"
  ) +
  theme_minimal()
```



Interpreting Each Coefficient: 1. Total Nitrogen (tn_ug_l = 6.661e-05)

Meaning: For every 1 µg/L increase in TN, $\log_{10}$(Chl-a) increases by 0.0000666 units Practical interpretation:

A 100 µg/L increase in TN → 0.00666 increase in $\log_{10}$(Chl-a) This equals 10^0.00666 = 1.015-fold change (1.5% increase in Chl-a)

Or more meaningfully: A 1000 µg/L increase in TN → 15% increase in Chl-a

2. Total Phosphorus (tp_ug_l = 3.883e-03)

Meaning: For every 1 µg/L increase in TP, $\log_{10}$(Chl-a) increases by 0.00388 units Practical interpretation:

A 10 µg/L increase in TP → 0.0388 increase in $\log_{10}$(Chl-a) This equals 10^0.0388 = 1.094-fold change (9.4% increase in Chl-a)

Strong effect: TP has much stronger per-unit effect than TN

3. Mean Depth (z_mean_m = −3.156e-02)

Meaning: For every 1 meter increase in depth, $\log_{10}$(Chl-a) decreases by 0.0316 units Practical interpretation:

A 5-meter deeper lake ⟶ 0.158 decrease in $\log_{10}$(Chl-a) This equals 10^-0.158 = 0.695-fold change (30% decrease in Chl-a)

Clear ecological meaning: Deeper lakes have less algae (light limitation)

4. Watershed Land Use (watershed_proportion = 2.524e-01)

Meaning: Lakes with high ag/urban watersheds have 0.252 higher $\log_{10}$(Chl-a) Practical interpretation:

High ag/urban watersheds ⟶ 10^0.252 = 1.79-fold higher Chl-a 79% more algae in agricultural/urban watersheds vs. natural watersheds

Model Performance:

$R^2 = 0.494$: Model explains 49% of variation in log(Chl-a) All predictors significant ($p < 0.001$) F-statistic highly significant: Overall model is valid

Ranking Effect Sizes:

Watershed land use: Strongest effect (79% difference) Lake depth: Moderate-strong effect (30% per 5m) Total phosphorus: Moderate effect per unit, but nutrients vary widely Total nitrogen: Weakest per-unit effect

Ecological Interpretation: This model shows that chlorophyll-a concentrations are primarily driven by:

Nutrient inputs (especially phosphorus) Light availability (deeper lakes = less algae) Watershed disturbance (agriculture/urban = more nutrients/algae)

The untransformed predictors make this easier to interpret in original units that managers understand!

A great write up

Results

Data Characteristics

The dataset comprised 593 lakes across Iowa, Wisconsin, and Michigan, with chlorophyll-a concentrations ranging from 0.5 to 384.2 μg/L (median = 8.7 μg/L). Total nitrogen ranged from 98 to 7,420 μg/L (median = 695 μg/L), while total phosphorus ranged from 3 to 410 μg/L (median = 25 μg/L). Lake areas varied from 0.01 to 647.5 km^2 (median = 0.4 km^2), with mean depths ranging from 0.3 to 21.3 m (median = 3.2 m). Watershed areas ranged from 0.1 to 9,847 km^2 (median = 23.8 km^2), and 47% of lakes had watersheds classified as high agricultural/urban land use (≥50% developed land).

Model Development and Transformation Analysis

Initial regression analysis using untransformed variables revealed severe violations of model assumptions, with significant heteroscedasticity (Breusch-Pagan test, $p < 0.001$) and non-normality of residuals (Shapiro-Wilk test, $p < 0.001$). Log_{10} transformation of the response variable (chlorophyll-a) substantially improved model diagnostics and assumption compliance.

Individual predictor transformation analysis using AIC comparisons showed that log transformations of total nitrogen (ΔAIC = 8.7) and total phosphorus (ΔAIC = 15.3) provided substantial model improvements, while transformations of other variables showed minimal benefit (ΔAIC < 2). However, to maintain interpretability in original measurement units, we retained untransformed predictors in the final model.

Multicollinearity assessment revealed acceptable variance inflation factors for all predictors (VIF < 2.5), indicating minimal redundancy among variables. Partial regression plots confirmed appropriate linear relationships between predictors and the log-transformed response variable.

Final Model Selection

Backward stepwise selection based on statistical significance ($\alpha = 0.05$) resulted in a final model containing four predictors: total nitrogen, total phosphorus, mean depth, and watershed land use proportion. Lake area and watershed area were removed during model selection due to non-significance ($p > 0.05$). The final model showed good fit to the data (adjusted $R^2 = 0.490$, $F_{4,588} = 143.3$, $p < 0.001$) and met all regression assumptions based on residual diagnostics.

Predictors of Lake Chlorophyll-a Concentrations

The final multiple regression model (Table 1) revealed that all retained predictors significantly influenced $\log_{10}$-transformed chlorophyll-a concentrations. Total phosphorus was the strongest predictor ($\beta = 0.00388$, $t = 12.51$, $p < 0.001$), followed by watershed land use ($\beta = 0.252$, $t = 6.57$, $p < 0.001$), mean depth ($\beta = -0.0316$, $t = -5.76$, $p < 0.001$), and total nitrogen ($\beta = 0.0000666$, $t = 5.07$, $p < 0.001$).

In terms of ecological significance, lakes with high agricultural/urban watershed land use had 79% higher chlorophyll-a concentrations compared to lakes in predominantly natural watersheds (antilog_{10} of 0.252 = 1.79). Each 10 µg/L increase in total phosphorus corresponded to a 9.4% increase in chlorophyll-a (antilog_{10} of 0.0388 = 1.094). Lake depth showed a strong negative relationship, with each additional meter of mean depth associated with a 3.1% decrease in chlorophyll-a concentrations (antilog_{10} of −0.0316 = 0.969). Total nitrogen showed the weakest per-unit effect, with each 100 µg/L increase corresponding to a 1.5% increase in chlorophyll-a.

Partial R^2 analysis (Table 2) indicated that total phosphorus explained the largest proportion of variance in chlorophyll-a (16.2%), followed by watershed land use (6.8%), mean depth (5.5%), and total nitrogen (4.2%). Together, these four variables explained 49% of the total variance in $\log_{10}$-transformed chlorophyll-a concentrations across the 593 study lakes.

Model Validation

Diagnostic plots confirmed adequate model performance (Figure 1). Residuals showed random scatter around zero with no systematic patterns, indicating appropriate model specification. The normal Q-Q plot revealed good adherence to normality assumptions, with only minor deviations in the tails. Scale-location plots showed relatively constant variance across fitted values, confirming homoscedasticity. Cook's distance analysis identified no highly influential observations (all values < 0.02), indicating robust model estimates.

Table 1. Multiple regression analysis of $\log_{10}$-transformed chlorophyll-a concentrations in 593 lakes across Iowa, Wisconsin, and Michigan.

Predictor	Coefficient	SE	t-value	p-value	95% CI
Intercept	0.588	0.036	16.29	<0.001	0.517, 0.659
Total nitrogen (µg/L)	6.66×10^{-5}	1.32×10^{-5}	5.07	<0.001	4.08×10^{-5}, 9.24×10^{-5}
Total phosphorus (µg/L)	0.00388	0.00031	12.51	<0.001	0.00327, 0.00449
Mean depth (m)	−0.0316	0.0055	−5.76	<0.001	−0.0424, −0.0208
Watershed proportion	0.252	0.038	6.57	<0.001	0.177, 0.327

Model: $F_{4,588} = 143.3$, $p < 0.001$; Adjusted $R^2 = 0.490$; Residual SE = 0.407

Table 2. Partial contributions of predictors to chlorophyll-a variance in the final multiple regression model.

Predictor	df	Sum of Squares	F-value	p-value	Partial R^2
Total phosphorus	1	25.67	156.5	<0.001	0.162
Watershed proportion	1	10.82	65.9	<0.001	0.068
Mean depth	1	8.76	53.4	<0.001	0.055
Total nitrogen	1	6.72	40.9	<0.001	0.042
Residuals	588	96.51			